Assignment 1 – Fall 2024

MA-110 Calculus and Analytical Geometry

CLO-1 , PLO-1

Q 1. **Compute** the domain and range of the following functions:

(a) $f(x) = 7x - 2$

(b) $f(t) = t^2 - 2 \quad 0 \leqslant t \leqslant 5$

(c) $g(x) = 3e^x + 4 \quad 0 \leqslant x \leqslant 2$

(d) $y(t) = (e^{2t} - 1)/2 \quad t \geqslant 0$

Q 2. **Sketch** the following functions:

(Label x- and y-axes and mention important values on the axes)

(a) $f(t) = \begin{cases} t & 0 \leqslant t \leqslant 3 \\ 3 & 3 < t \leqslant 4 \end{cases}$

(b) $g(x) = \begin{cases} 2 - x & 0 \leqslant x < 1 \\ 2 & 1 \leqslant x \leqslant 3 \end{cases}$

(c) $a(t) = \begin{cases} 1 - t & 0 \leqslant t \leqslant 1 \\ t - 1 & 1 < t \leqslant 2 \end{cases}$

(d) $b(x) = \begin{cases} 2 & 0 \leqslant x \leqslant 1 \\ 1 & 1 < x \leqslant 2 \\ 3 - x & 2 < x \leqslant 3 \end{cases}$

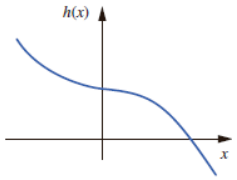
(e) $f(x) = x^3 + 8$

Q 3. Given $f(t) = t^2 + 1$, $g(t) = 3t + 2$ and $h(t) = \frac{1}{t}$, **express** the given functions in the simplest form:

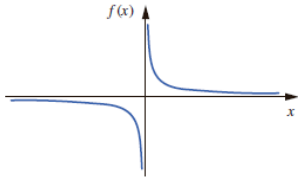
(a) $f(h(t))$ (b) $f(g(t))$

Q 4. **Classify** the following functions as even , odd or none.

(a)



(b)



(c) $y(x) = \dfrac{3x + 1}{x - 2}$

(d) $h(s) = \dfrac{9}{(s + 2)(s - 1)}$

To be submitted latest by 7th October 0920 hrs

Note: Your submitted assignment must be your own effort. Copying from your fellow or internet or any other source may lead you to F-grade in the course. Anyone who shares his/her solution with fellows will get the same penalty.